

Electric Potential (completed)

Goal.

- See electric potential as charge's version of *height*: PE per charge, not PE itself.
- Ask whether V stays constant across a region where $\vec{E}$ is constant (a parallel-plate capacitor) — and check that against the same question for gravity.
- Convert between a potential difference and a change in speed (Example 25.6).
- Build V for one point charge, then for several by *adding numbers*, not vectors (Example 25.9).

1. Electric potential is charge's “height”

A ball's gravitational PE isn't just a property of the ball — you need two more things:

- The mass of the object.
- Height *relative to where* — height means nothing until you pick a reference (ground level, table height, ...).

Electric potential energy works the same way. You need:

- The amount of charge.
- Potential *relative to where* — same requirement, a reference has to be chosen.

Electric potential is potential energy *per charge* — the part of that pair that doesn't depend on which charge you put there:

$$V \equiv \frac{U}{q}$$

measured in **volts (V)**, where $1 \text{ V} = 1 \text{ J/C}$.

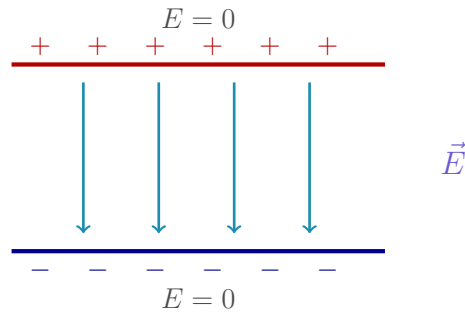
Height tells you PE *per mass*. Potential tells you PE *per charge*. Neither one is a force, and neither one is an energy by itself — both need the “per” filled in with an actual mass or charge before they turn into PE.

Check. You're told $V = 50 \text{ V}$ at some point. Can you find the PE a charge would have there? What's missing?

Not yet — V alone is PE per charge, not PE. You need the actual amount of charge q you'd put there: $U = qV$.

2. Does V stay constant where $\vec{E}$ is constant?

A parallel-plate capacitor: two sheets of charge, surface density $\pm\eta$, close together.

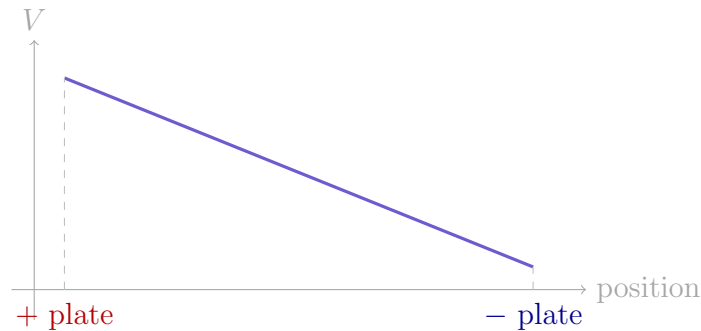


- Each plate alone gives a field of magnitude $\frac{\eta}{2\epsilon_0}$ on either side of it.
- Between the plates, the two fields point the **same** direction and add: $E_{\text{total}} = \frac{\eta}{\epsilon_0}$ — uniform, pointing from $+$ to $-$. Outside the plates they **cancel**: $E = 0$.

Check. $\vec{E}$ is exactly constant everywhere between the plates. Does that mean V is constant there too? Compare to gravity: is gravitational PE constant wherever the *force* of gravity is constant?

No, on both counts. $U = mgh$ keeps changing with h even though g (and the force mg) never changes near Earth's surface. Same here: V keeps changing steadily as you cross the gap, even though $\vec{E}$ itself never does.

Sketch V as you walk from the $+$ plate to the $-$ plate:



- The $+$ plate is at the higher potential.
- $\vec{E}$ points from the $+$ plate to the $-$ plate — i.e. toward **decreasing** V . (Same rule as always: $\vec{E}$ points “downhill.”)

3. Worked example: crossing a potential difference

A proton with speed 2.0×10^5 m/s enters a region where it crosses a potential difference of 100 V. What's its final speed? What if it were an electron instead?

Energy conservation, with $U = qV$:

$$\frac{1}{2}mv_f^2 = \frac{1}{2}mv_i^2 - q\Delta V \quad \Rightarrow \quad v_f = \sqrt{v_i^2 - \frac{2q}{m}\Delta V}$$

Proton ($q = +e$, $m = 1.67 \times 10^{-27}$ kg):

$$(v_f)_p = \sqrt{(2.0 \times 10^5)^2 - \frac{2(1.60 \times 10^{-19})(100)}{1.67 \times 10^{-27}}} = 1.4 \times 10^5 \text{ m/s}$$

Electron ($q = -e$, much smaller m):

$$(v_f)_e = 5.9 \times 10^6 \text{ m/s}$$

Check. The proton slowed down; the electron sped up. Both crossed the *same* potential difference. Why the opposite result?

The sign of q flips the sign of $q\Delta V$ in $v_f^2 = v_i^2 - \frac{2q}{m}\Delta V$. A positive charge moving toward higher V gains PE and loses KE (slows); a negative charge moving the same direction *loses* PE and gains KE (speeds up) — exactly the $+q/-q$ split from the uniform-field pushing argument two classes ago.

4. Potential of a single point charge

Start from the PE of two point charges, and divide out the second charge — the same move that turned U into V up in §1.

$$U_{12} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r_{12}} \quad \Rightarrow \quad V_1 = \frac{U_{12}}{q_2} = \frac{1}{4\pi\epsilon_0} \frac{q_1}{r_{12}}$$

Given the potential of a point charge, what do you still need to get an actual PE out of it?

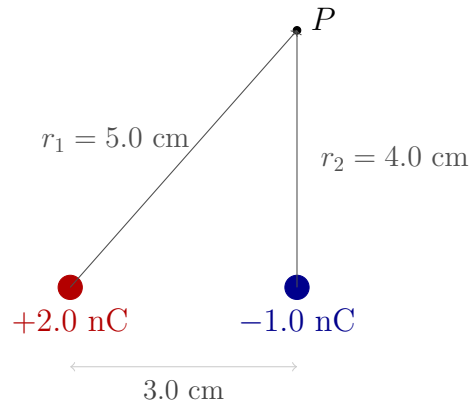
- The amount of the *other* charge you'd place there.
- **Nothing else** — a point charge's own potential already comes with a reference built in: $V \rightarrow 0$ as $r \rightarrow \infty$.

Check. As $r \rightarrow \infty$, what does V_1 approach? Does the *sign* of q_1 matter to V_1 ?

$V_1 \rightarrow 0$. Yes — unlike $|E|$, V_1 keeps the sign of q_1 : positive charges make positive potential everywhere around them, negative charges make negative potential.

5. Potential from several charges: just add the numbers

Example 25.9. A $+2.0 \text{ nC}$ charge is 5.0 cm from point P ; a -1.0 nC charge is 4.0 cm from P . Find V at P .



The potential is a *scalar* — no angles, no components. Just add the two numbers:

$$V = V_1 + V_2 = \frac{1}{4\pi\epsilon_0} \frac{q_1}{r_1} + \frac{1}{4\pi\epsilon_0} \frac{q_2}{r_2}$$

$$V_1 = (8.99 \times 10^9) \frac{2.0 \times 10^{-9}}{0.050} \approx +360 \text{ V} \quad V_2 = (8.99 \times 10^9) \frac{-1.0 \times 10^{-9}}{0.040} \approx -225 \text{ V}$$

$$V = +360 - 225 = +135 \text{ V}$$

Check. Suppose you needed $\vec{E}$ at P instead of V . Would this same “just add the numbers” shortcut work?

No — $\vec{E}$ is a vector, so adding the two charges’ fields needs their directions (components or angles), not just magnitudes. That’s the entire payoff of working with V instead of $\vec{E}$ whenever you only need energy: no geometry to track.